

National College of Ireland

Project Submission Sheet

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Programme: Msc in Data Analytics **Year:** 2026
Module: Deep Learning and Generative AI
Lecturer: Dr Mohamed Khalafalla Hassan
Submission Due Date: 20-08-2026
Project Title: Examinations (TAB A)
Word Count: 4500

Academic Integrity Certificate Verification Code: 6IUoUeD6C8

I hereby certify that the information contained in this (my submission) is information pertaining to research I conducted for this project. All information other than my own contribution will be fully referenced and listed in the relevant bibliography section at the rear of the project.

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Signature: Subhash
Date: 15-08-2026

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AI Acknowledgement Supplement

Deep Learning and Generative AI

TABA exam

Your Name/Student Number	Course	Date
Subhash Lora Jat / 25171496	Msc in Data Analytics	15-08-2026

This section is a supplement to the main assignment, to be used if AI was used in any capacity in the creation of your assignment; if you have queries about how to do this, please contact your lecturer. For an example of how to fill these sections out, please click [here](#).

AI Acknowledgment

This section acknowledges the AI tools that were utilized in the process of completing this assignment.

Tool Name	Brief Description	Link to tool
Gemini	Used for comparison, as per question requirements	Gemini

Description of AI Usage

This section provides a more detailed description of how the AI tools were used in the assignment. It includes information about the prompts given to the AI tool, the responses received, and how these responses were utilized or modified in the assignment. **One table should be used for each tool used.**

[Insert Tool Name]
Gemini
I am a DL Analyst/Engineer building a Retrieval-Augmented Generation (RAG) system for domain-specific question answering over a college's internal knowledge base (admissions policy, fees, registration, library referencing guidance, exam/results FAQs). Propose a methodology covering: data preparation, choice of modelling techniques, hyperparameter optimization, model evaluation, scalability, emerging technologies to consider, and ethical/responsible AI considerations.

Evidence of AI Usage

This section includes evidence of significant prompts and responses used or generated through the AI tool. It should provide a clear understanding of the extent to which the AI tool was used in the assignment. Evidence may be attached via screenshots or text.

Additional Evidence:

[Upgrade](#)

I am a DL Analyst/Engineer building a Retrieval-Augmented Generation (RAG) system for domain-specific question answering over a college's internal knowledge base (admissions policy, fees, registration, library referencing)

Data Preparation & Ingestion Pipeline

Raw Documents (PDF, DOCX, HTML, FAQs)



[Layout-Aware Extraction] → [Structural & Security Metadata Injection]

+ Ask Gemini

Flash



Gemini is AI and can make mistakes, including about people. [Your privacy and Gemini](#)

emerging technologies to consider, and ethical/responsible AI considerations.

[Upgrade](#)

Data Preparation & Ingestion Pipeline

Raw Documents (PDF, DOCX, HTML, FAQs)



[Layout-Aware Extraction] → [Structural & Security Metadata Injection]



[Hybrid Chunking Strategy] (Hierarchical + Contextual Retrieval Wrapping)

+ Ask Gemini

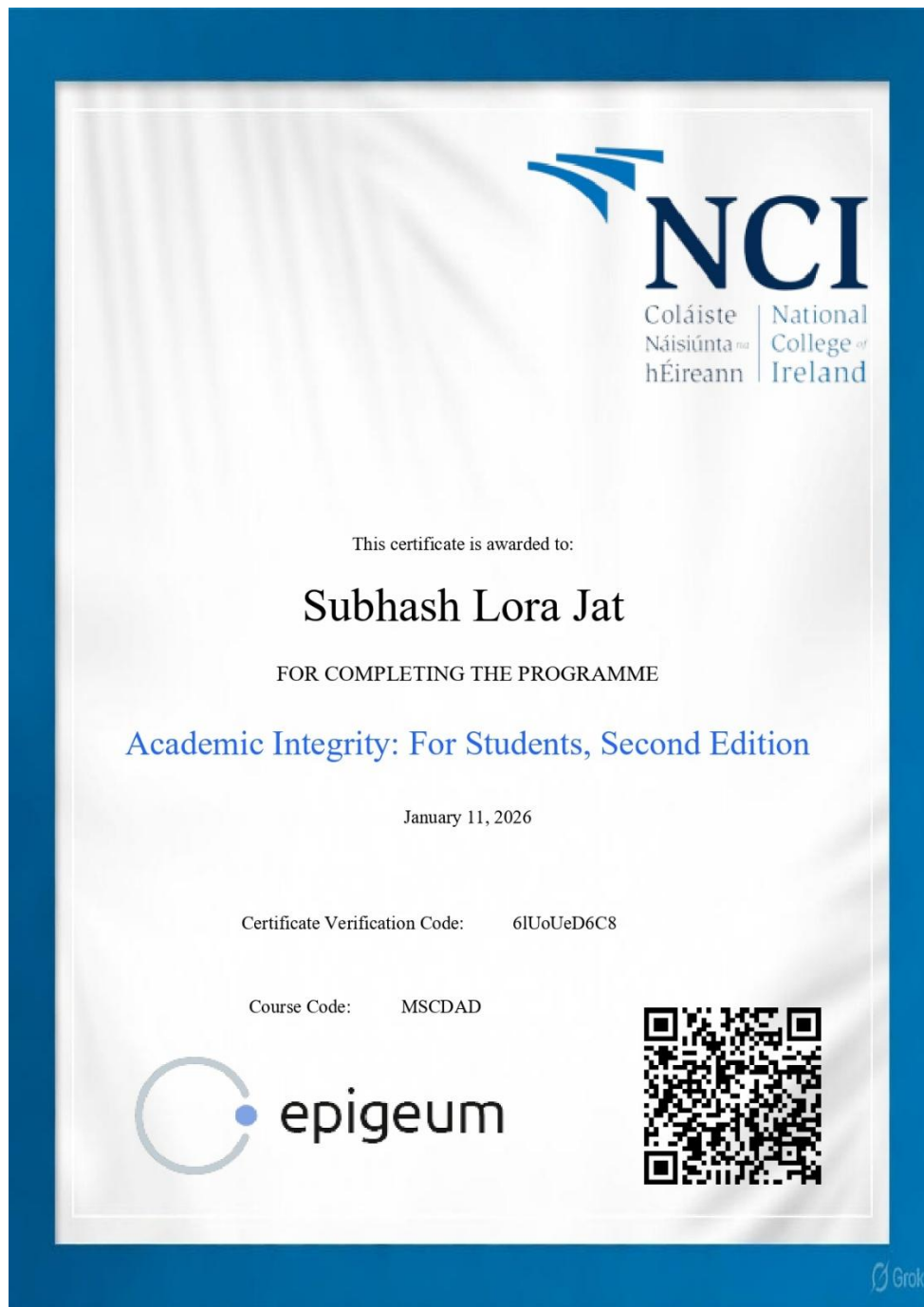
Flash



Gemini is AI and can make mistakes, including about people. [Your privacy and Gemini](#)

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<https://libguides.ncirl.ie/academicintegrity/epigeumforstudents>
**Completion of Epigeum Academic Integrity Online Course is Mandatory for
all registered students of NCI**

Q1 : A Hybrid Retrieval-Augmented Generation Methodology for Domain-Specific Question Answering

Introduction and Assumptions

This report is based on a methodology for RAG system for domain specific question answering we have developed a hybrid version of 2 recently published approaches. These papers are retrieval aware fine tuning framework for an in house product QA system published by (sharma et al 2024) and other is BM25/dense retrieval system used for an institutional question answering an case study (Sun, Wang, and Zhang, in 2024). This system is evaluated using NCI documents which are public to students like what is the fees , plagiarisms , this info is used as a domain specific. This report is a working proof of concept.

DATA PREPARATION

For proof of concept we gathered the documents from the NCI's publicly available information, it included different categories information which must be answered by the NCI assistant, these are Registration booklet which include fees, enrolment , orientation, campus life) , Handbooks, NCI library referencing and Avoiding the Plagiarism guide, School of business results FAQ and last student card application guide. When we added up , these were total 30,976 words across all the documents which means 6,195 words per document. This reflects the heterogeneity of the institutional document.

The data was taken from HTML and Pdf pages , also it was cleaned slightly like removing the extra spaces , removing the repeating boilerplate. We intensively did not done the heavy cleaning as it would remove very important information like fee tables , numbered steps , deadlines etc.

Then splitted text into chunks using window method. Chunk size was 150 words after doing testing and tuning. Also with 30 words overlap, total chunks were 260. All processing was running smoothly on single machine without using GPU.

Also there is need to add the pipeline if we add the personal data , as if we extend the beyond the current five public documents, followed by Sharma et al (2024).

CHOICES OF MODELLING TECHNIQUES

This methodology splitted RAG into 2 parts retriever and generator, then it makes a separate design choices for each. We have selected these choices on strengths and weaknesses of the paper A (Adobe system by Sharma et al 2024) and paper B (CMU/Pittsburgh system by Sun , Wang and Zhang published in 2024).

Retriever

The retriever was combined by three parts, instead of using the click log data like paper A we used a dense embedding retriever which was lightly fine tuned using weak supervision signals. Then a BM25 sparse retriever was used from paper B , then we applied a lightweight reranker on the combined results from Paper B.

This combination was chosen because each paper does only half of it: paper A fine tunes a dense retriever but uses no sparse retrieval, while paper B combines sparse and dense retrieval with FAISS and reranking but does no fine tuning. I observed that both of the papers haven't used fine tune dance retriever with sparse retrieval and re ranking together and that was the gap exactly this project is trying to fill.

We have chosen RRF (Cormack, Clarke and Buettcher, 2009) to combine the BM 25 and the dense results instead of adding the weighted scores together. The reason was BM 25 endance course were completely on a different scales, if we add their weights it would risk the one retriever over the other, RRF has helped by combining on basis of rank position instead of raw scores.

We have seen that Paper A results have scored their fine tuned generator lower than the simple retriever with GPT 4 setup. Same findings were in this project which confirms that dense only underperformed the hybrid retrieval on one query. We also observed that BM25 Retriever was also rejected, although it has performed very well there , but the core reason was that our BM 25 Was struggling with paraphrased questions but were not matching exact words from the source text.

Generator

There was a already a ready made instruction tuned LLM with retrieval augmented prompt used by the generator. This rejects paper A's approach of fully fine tuning the generator on negative document pairs, which is weakly supported without Adobe's internal training data and needs computing resources and curated data that were not available in the time given.

Paper B approach which uses a frozen model with few prompts was followed, A smaller instruction tuned model replaced a self hosted 7B model, reducing infrastructure without affecting retrieval. In the future if the proper NCI specific training data will be available then a full generator fine tuning work will be implemented.

HYPERPARAMETER OPTIMISATION

There were a 22 hand labelled queries were tested against 2 RAG specific settings which covers all the 5 source documents with a chunk size overlap and the RRF fusion constant were used to combine both BM25 and dense retrieval results. The measurement of the results was taken by the dense encoder, all miniLM L6 v2 (Reimers and Gurevych, 2019).

Chunk Size and Overlap

The combinations of small chunk size and overlap were tested these are (100, 20), (150, 30), (200, 40). Each of the junk sizes was covering around 20% of the chunk size, this is a default. We have also tested 150,0 to see that what would happen if we remove the overlapped completely.

We have chosen the chunk size of 150 and overlap with 30, because it was the smallest configuration where both hybrid retriever and BM25 reach at a perfect recall and the MRR score of 1.0/1.0. The other option of 200, 40 was also matching the hybrid performance but it left the BM 25 noticeably weaker where MRR of 0.955 Instead of 1.0, which makes it a less reliable choice.

When we remove the overlap it had made the condition more worse, our BM 25 recall was dropped from 1.0 to 0.955 and dance was dropped from 0.955 to 0.909. These numbers show that how important is the overlap because practically it helps in stopping the relevant text from being cut off add a junk boundaries, this sensitivity was noted by the Sun, Wand, Zhang (2024)

RRF Fusion Constant

We have tested the RRF fusion constant at the best chunk configuration of 10, 60 and 100. All these 3 have given the same perfect recall and MRR of 1.0/1.0. This may be due to the small size of the corpus with only 260 chunks be the different vocabulary in every document, that's the reason of getting a better results regardless of which fusion constraint is used.

As this projects scale is not able to meaningfully tell the difference between the 3 options so the extended value of 60 from the literature (Cormack, Clarke and Buettcher, 2009) was kept as a safe and a well established choice. When the corpus will grow larger the sensitivity to the fusion constant will also tested once again rain queries become more varied.

Fine-tuning Hyperparameters

We have also not done the fine tuning settings for the retriever for example learning rate, bed size and contrastive temperature. The reason was that current version was using the pre trained encoder without using the fine tuning step of the weak supervision this step was the methodologies original post step. Due to resource limitation of a 2 weeks time it was not an intensively designed decision.

On the basis of find tuning sentence embedding models which has a contrast objective (Reimers and Gurevych, 2019), some has suggested that our starting values for our future step must be from $2e-5$, with a batch size of 16 to 32, along with the temperature of around 0.05 and 3 to 5 training epochs With stopping based on validation set. When weak supervision relevance labels are collected these have to be tuned properly once again.

MODEL EVALUATION

We have measured the retrieval quality by using the all 22 labelled queries where we just do the relevance only at the level of the documents, we have counted the chunk as a correct only when it has come from the document from which it was written. We have used recall@3 and Mean reciprocal rank (MRR) instead of using nDCG, the reason for not using ndcz was that it needs a grade, multi level relevance scores. The projects like this are very small so they are most likely not to produce these type of results although nDCG is the better metric for a full production system. The Sharma et al 2024 has used this system in a detailed for the better judgments.

This paper has compared a 3 retrieval steps these are BM25 only, dense only and the proposed hybrid, This hybrid approach combines both BM 25 and dense rankings by using RRF following lexical overlap re ranking.

First Evaluation Round — Substitute Encoder

In the first time, we used the substitute dense encoder as downloading the full sentence embedding was not possible in the environment. In fact, TF IDF vectors was used into a fixed dense space using the random projection. When we compared the results of this setup, we get BM25 rescall@3/MRR were 1.0/1.0, whereas dense only performed 0.864/0.758 and our hybrid get 1.0/0.947. here hybrid has scored lower than the MRR and was not able to win over the BM25, during the rank fusion weak dense signal was pushing the correct chunk down occasionally. This was substitute encoder's limitation not the hybrid design itself.

Second Evaluation Round — Intended Encoder

After installing the intended encoder we have repeated the evaluation all MiniLM L6 v2 (Reimers and Gurevych, 2019). The results were improved very good we have seen that BM25 has remained add 1.0/1.0, and dense has rose up to 0.955/0.932. Our hybrid scored 1.0/1.0 which was matching perfectly.

By looking closely to the individual queries we notice that why did the hybrid has earned this result then just tying to the BM 25. The query was "what transport options are available from getting to NCI?", our dense retriever return the chunks from the wrong document quality assurance handbook. This document was the largest document in the corpus and it looks that dense retriever was pulling towards it. Whereas when we look towards BM 25 on the other hand it has correctly matched the exact terms from the right source for example "transport", "Luas" and "Dublin Bus". A rank fusion has recovered the correct chunk at the rank one but the dense retriever has missed on that query. This is the evidence that combining both retrievers will add the robustness neither one provides on its own.

Process Finding

We have also added one additional thing for only transparency purpose the earlier version of lexical overlap was Continuously overlapping the term by using all the tokens including stop words also, this has caused it to favour the long chunks which shared a common words with the query, whereas shorter chunks were neglected which contains the actual correct answers. We have find it while inspecting the failure in the retrieval during our development and fixed it by limiting the overlap scoring to contained words only. The results shown above are the corrected versions.

Limitations

With a small amount of documents (five) with 260 chunks, our BM 25 has alone get the high performance, due to the small keyword heavy corpus, That's the reason our hybrid retrieval has a no chance to show the advantage over the BM 25. The transport query has shown that this advantage would be bigger and more obvious on a larger and varied corpus as we have seen in the results of the transport query, where BM 25's exact match weakness was seem often. This remains as an assumption at a production level not a proven result.

Among the generation side like covering the faithfulness, answer relevancy and context precision also using the recall RAGAS style framework was not carried out, because no generator component was actually built in this project retrieval was our main focus within the available time.

SCALABILITY ISSUES

The BM 25 and the indexes in this project are built in the memory with only 260 chunks and it answers within the 5 milliseconds. This is not representing that how full nci wide deployment would be look like there are 3 scalability concerns which we will discuss now.

Vector Store Growth

NCI full wide corpus would likely to reach around thousands of chunks. At that level the approach brute force cosine similarity search would become very slow, at that level we need a nearest neighbour index like FAISS , which is used by the Sun, Wang and Zhang (2024) , Would be needed to keep our dense retrieval fast for the practical

Incremental Re-indexing

As I am a student of the NCI , I noticed that the policy documents of the college do not change often, so for that a scheduled full re index once a week would be a simple and low cost solution, rather than implementing a real time updates. This is really different from the Adobe's situation described in Sharm et al 2024 , where products documentation change is very frequently and it would need a different approaches at a time.

Reranker Cost

The lexical overlap re ranker which we have used in this project is really very cheap to run, where the miniLM cross encoder it stands in , which is used in the paper B Requires a separate pass for each candidate chunk. When it is done at larger level, this will add up , for that batching re ranker calls and the number of candidates considered would be likely necessary to keep the responses times under the control which we have already done here by using a fixed top K

EMERGING TECHNOLOGIES

Newer published development reports after the 2 sources directly relevant to extending this project though one of them was used in this 2 week proof of concept.

RAFT — Retrieval Augmented Fine-Tuning

RAFT was also introduced by the Zhan et al (2024), in this concept it rains the generator to tell the difference between useful documents and irrelevant documents that get retrieved alongside them. This is done by using chain of thoughts style of answers which directly cites the passages of the source. The gap which we left for the future in this project this is filled by this RAFT approach. Instead of fine tuning the grounded and the negative pairs as the paper a has done, RAFT is offering more recent and RAG specific choice that could be applied to the NCI question answer training data.

GraphRAG

Another technology proposed by the Edge et al 2024 named as Graph RAG, in this approach it builds a graph of the knowledge at the entity level and generates the summary descriptions for clusters of the related content of the corpus. This helps us to answer the wide questions like "what are all the deadlines this semester ? " These kind of questions are harder to be answered by the normal retrieval, although no single chunk contains the full answer. Most of the queries which we have answered are based on the single fact lookups which are easy to handle. But on a broader concept NCI assistant will have to handle wider questions that may have a multiple documents. This Graph RAG approach of summarization could be a option along with our proposed hybrid retriever for those kind of queries , not a replacement for it.

Long-Context LLMs

Nowadays , new LLMs supports the context windows of hundreds of thousands of tokens, this thing raise the question that whether we need retrieval , when a whole corpus is used in the context. This is technically possible in this project as it contains around 31,000 words. But it will not remove the value of retrieval with the context length on single query, it doesn't matter how relevant the content is the original RAG paper by Lewis et al 2020. It motivates to keep only precise and only relevant and up to date information in the context rather than depending on what model already stored in its parameters. The reasoning behind this becomes more important , when NCI's corpus grows larger, as it fits in the single window at productions.

Adaptive and Self-Querying Retrieval

In the adaptive retrieval system here the system decides whether the retrieval is needed or not, also it reformulates the query before searching was also not implemented in this project. But it would be a very useful extension in handling the conversations for the follow up questions. This helps in understanding the users actual meaning like what user is mean to search is different from what they literally typed. This would be our natural next step once our system moves beyond the single turn question answering.

ETHICAL, LEGAL AND RESPONSIBLE AI CONSIDERATIONS

The Ethical, legal and responsible AI considerations had came from findings from this project rather than explaining from the general way.

PII Handling

The documents which I have used in this project does not contain any of the personal information. In future if there is any requirement of adding , then surely this project would implement the proper PII removal step as used by the Sharma et al 2024 before indexing. This is an crucial step as chunks are created directly from the data , and its results are directly shown to the users, so any personal information left in the text would directly risk at retrieval time, not just during training. Such things first must align with GDPR, more around right to erasure and right to access.

Content Ownership

As we know that the public documents available on the website are institution property, therefore any retrieval system which is built on it must used as the internal NCI tool, it cannot share there results outside. This limits on the integrations of NCI with the third party generative AI api's.

Bias

The findings which we recorded in the model evaluation, as we discussed in that section that dense retriever displayed the largest document in the corpus on single query , where it choose the larger instead of the smaller which was the correct one, this was not just an accuracy issue also it seems as a fairness issue. If we do not keep the BM25 in our hybrid approach , then in future when document production will scaled at larger level then this biasness issue will surely maximise.

Hallucination

As this is well known that retrieval grounding does help in reducing the hallucinations , but it does not remove it completely. However we havn't used any generator in this project , but when we compared our methodology with the general purpose AI we get the mitigation that prompt generated must be clearly instruct the model to answer the retrieved context and also show when the answer of the context is not present, by this there would be less auto generated responses rather than completely rely on the retrieval quality.

Responsible AI Communication

As we are already doing the project on the NCI , and it already reflects its guidance published on the responsible use of the generative AI by students (Mallon, Brittle , and Dunne 2025). This will help in shaping the NCI assistant to how should be shown to students, making clear that it is an AI system not any human authority.

COMPARISION WITH GEN AI

Note: This part of the report is excluded from the word count as per requirements.

Prompt Submitted

The following prompt was submitted to Gemini (Google) on 12 August 2026:

"I am a DL Analyst/Engineer building a Retrieval-Augmented Generation (RAG) system for domain-specific question answering over a college's internal knowledge base (admissions policy, fees, registration, library referencing guidance, exam/results FAQs). Propose a methodology covering: data preparation, choice of modelling techniques, hyperparameter optimization, model evaluation, scalability, emerging technologies to consider, and ethical/responsible AI considerations."

The full response is included in the Appendix. A summary of the comparison follows.

Similarities

The similarities identified while comparison are listed below:

- Dense embedding was combined with sparse BM25 retrieval
- With same k value of 60 followed by reranking stage , it fused by using the RRF
- They were also treating the hyperparameters tune against metric rather than fix by assumptions.
- Gemini has named the approaches which we have tested in our project like chunk size , dense/sparse weighting , reranker cutoff.
- Also suggested the same hallucination mitigation.

Differences

We have seen that Gemini has written its response at the production level. It has proposed an hierarchical parent child chunking contextual retrieval, semantic caching , event driven Kafka-based re indexing, an agentic query router, components like Corrective Rag and Self Rag and knowledge graph integration for multi hop questions. This was a large proposal its not possible to complete in two week project deadlines. Its more like a blueprint for an well enterprise .

It was not citing any literature as it was showing the commercial and open source tool like Cohere Rerank instead of research papers. Here this report has a proper citations from the peer reviewed literature.

Our gemini model was also talking about the numeric thresholds, like blocking the deployment in case the faithfulness falls below 0.95 or context recall below the 0.88 without any justification like from where does the

numbers came from with no evidence. But here in this reports no numbers were mentioned without any proof , we mentioned every results with proof.

Strengths and Limitations of Both

Its strength is that it produces the real engineering breadth, as it has raised the concerns related to operations in depth whereas our report lag this , like role based access control and meta based security filtering. But its drawback is that its not tested with any real constraints. As it has proposed the Kafka pipelines and graph based owner knowledge infrastructure, whose corpus is 5 documents and 260 chunks which does not engage any kind of resourcing reality.

This report is although looks very narrower but it has cited its literature and most importantly with a measured results in a working system but its limitation is that, it's proof of concept is much more smaller.

Did the Gemini Response Add Any Insight?

Sure Gemini has added an very useful insights for instance it has covered the role based access control and metadata drive and security which filtering public on the basis of student and staff. But our current proposal is entirely based on a public NCA documents so it does not need any control at this scope. However we know that Gemini is correct in this aspect there must be a deployment of internal policy drafts or some confidential procedures alongside public pages, we have added this have the scope in the previous sections.

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APPENDIX : FULL GENERATED AI RESPONSE – GEMINI

Data Preparations and Pipeline

Pipeline: Raw Documents (PDF, DOCX, HTML, FAQs) -> Layout-Aware Extraction -> Structural & Security Metadata Injection -> Hybrid Chunking Strategy (Hierarchical + Contextual Retrieval Wrapping) -> Dual Vector/Lexical Indexing (Dense Embeddings + BM25 Sparse Index).

Document Ingestion & Parsing - Layout-Aware Extraction: College knowledge bases contain complex PDF tables (fee structures, exam timetables) and multi-column policy documents. Standard text extractors destroy tabular layouts. Use layout-aware parsers (e.g., Unstructured, LlamaParse, or Microsoft MarkItDown) to parse document hierarchies into clean Markdown, preserving tables as HTML/Markdown blocks. OCR & Document Normalization: Run OCR (e.g., PaddleOCR, Tesseract) on scanned historical policy documents and standard forms.

Structural & Security Metadata Injection - Pre-pend every extracted document with structured metadata fields prior to chunking: document_type (policy, faq, curriculum_guide, fee_schedule); target_audience (undergraduate,

postgraduate, international, staff)); effective_academic_year (2025-2026, 2026-2027); access_level (public, student_authenticated, staff_only); source_url and section_path (e.g. /registrar/policies/deferral#section-3).

Hybrid Chunking Strategy - Hierarchical (Parent-Child) Chunking: Split source documents into large parent chunks (e.g., 1,524-2,048 tokens) for LLM context, and smaller child chunks (e.g., 256-512 tokens with 15% overlap) for vector indexing. Retrieval matches precise child chunks but returns the broader parent chunk to the generator to maintain context integrity. Contextual Retrieval Injection: Prepend each chunk with a brief 2-3 sentence LLM-generated summary detailing where the chunk fits within the broader document (e.g. 'This chunk is from the 2026 Academic Code of Conduct, Section 4: Examination Misconduct'), reducing context isolation.

Choice of Modelling Techniques

Pipeline: User Query -> Query Deconstruction & Expansion -> [Dense Vector (bge-m3 / E5) and Sparse BM25 (Exact Terms/IDs)] -> Reciprocal Rank Fusion -> Cross-Encoder Reranker (Cohere / bge-reranker) -> Instruction LLM Generator.

Hybrid Retrieval Architecture: RAG applications for academic regulations require both broad semantic mapping and exact keyword matching (e.g., course codes like CS8012, policy numbers, fee codes). Dense Semantic Search: use high-performing embedding models optimised for retrieval (e.g., bge-m3, text-embedding-3-large, or e5-mistral-7b). Sparse Lexical Search: implement BM25 or SPLADE for term-exact matching on administrative terminology and alphanumeric identifiers. Fusion Layer: merge rank lists using Reciprocal Rank Fusion, $RRF(d) = \text{sum over retrieval systems } m \text{ of } 1/(k + r_m(d))$, where $r_m(d)$ is the document's rank in system m and k is a smoothing constant, typically $k=60$.

Cross-Encoder Reranking: initial retrieval fetches the top $K=50$ candidates across dense and sparse retrievers; pass (query, chunk) pairs through a high-precision cross-encoder reranker (e.g., bge-reranker-v2-m3 or Cohere Rerank 3.5) to output a unified relevance score, narrowing context down to the top $N=5$ most relevant chunks.

Generator Architecture: deploy models fine-tuned for instruction following and complex context synthesis (e.g., Llama-3.1-70B-Instruct, Qwen-2.5-72B-Instruct, or GPT-4o / Claude 3.5 Sonnet via API). Prompt Enforcement: force exact inline citations mapping to retrieved chunk IDs; require explicit refusal ('I cannot find this information in the official policy') when retrieved context confidence falls below threshold.

Hyperparameter Optimisations (HPO)

To maximise performance, use an automated HPO framework (e.g., Optuna or Ray Tune) across the core pipeline configuration space:

Component	Parameter	Search Space	Optimization Metric
Chunking	Child Chunk Size	128-768 tokens	Retrieval Recall@K / Context Relevance
Chunking	Chunk Overlap	0%-20%	Information Coverage
Hybrid Search	Dense Weight (alpha)	0.0-1.0	MRR@10
Reranking	Initial Candidate Set (K)	20-100 chunks	NDCG@5 vs. latency
Reranking	Context Cutoff (N)	3-10 chunks	Generator Hallucination Rate
Generation	LLM Temperature	0.0-0.2	Faithfulness / Groundedness

Model & System framework

Implement a multi-tiered evaluation harness leveraging 'The RAG Triad' alongside classical retrieval metrics. Retrieval Quality: MRR@K (speed of finding the first relevant document); NDCG@K (quality of ranking relevant chunks near the top); Context Recall & Precision via Ragas/TruLens (proportion of ground-truth evidence retrieved vs. proportion of retrieved chunks actually relevant). Generation Quality: Faithfulness/Groundedness (whether claims are strictly derived from retrieved context - hallucination detection); Answer Relevancy (how directly the generated text answers the question).

Evaluation Execution Workflow: (1) Golden Dataset Creation - collaborate with university administrators to curate 300+ ground-truth Q&A pairs with verified source links, augmented with synthetic query generation (e.g., Ragas). (2) Automated CI/CD Evaluation Gate - integrate Ragas, TruLens or DeepEval into the build pipeline; block deployments if Faithfulness falls below 0.95 or Context Recall drops below 0.88. (3) Production Observability & Monitoring - track user feedback (thumbs up/down, edit requests) and log low-confidence retrievals to identify knowledge-base gaps.

Scalability and Operational infrastructure

Pipeline: User Request -> API Gateway & RBAC -> Semantic Caching (Redis) -> on cache hit, instant answer; on cache miss, Hybrid Retrieval Pipeline -> Vector DB (Qdrant/Milvus).

Vector Database Infrastructure: deploy distributed vector stores (e.g., Qdrant, Milvus, or Pgvector) using HNSW indexing for sub-20ms search latencies over millions of vectors. Metadata & Payload Filtering: pre-filter at the index layer to enforce metadata security boundaries (e.g., access_level <= user_role) prior to similarity calculation. Semantic Caching: place a semantic cache (e.g., Redis VL or GPTCache) ahead of the retrieval pipeline using cosine similarity threshold >= 0.96, so repeated queries serve immediately without triggering LLM calls. Event-Driven Knowledge Ingestion: implement change-data-capture (e.g., Apache Kafka / RabbitMQ) connecting CMS/drive platforms to vector ingestion, re-indexing updated policy files incrementally without full database re-indexing.

Emerging technologies and Advanced patterns

Agentic RAG: replace static pipelines with an LLM router agent that classifies the query and routes to specific workflows (e.g., a fee calculation query to a Python execution tool, an admissions policy query to vector search, a library search to an API). Corrective RAG (CRAG) & Self-RAG: integrate lightweight evaluator components that score retrieved chunks; if context relevancy falls below a threshold, trigger query expansion or a human-in-the-loop fallback. GraphRAG (Knowledge Graph Integration): combine vector embeddings with knowledge graphs (e.g., Neo4j) to support multi-hop relational questions (e.g., 'If I defer my admission due to medical reasons, what impact does it have on my prerequisite eligibility for CS201 and my scholarship status?').

Ethical & responsible AI considerations

Role-Based Access Control (RBAC) & Data Isolation: ensure strict tenant and user-level isolation - undergraduate student queries must never retrieve administrative staff policy notes, confidential grade dispute logs, or unannounced policy drafts. Strict Hallucination Prevention & Abstention: enforce system guardrails (e.g., NeMo Guardrails or Llama Guard); if retrieval confidence is low, the model must explicitly respond 'I cannot find an official college policy answering this question. Please contact the Registrar's office directly.' Source Traceability & Auditability: every output claim must contain traceable citations specifying document title, exact section, and version date. Bias & Fairness Mitigation: ensure policies on international student fees, visa extensions, and disability accommodations are retrieved neutrally without bias introduced by generative re-paraphrasing.

Q2: Review of the Cambrian-1

STRUCTURE, LANGUAGE AND TITLE

The paper which we reviewed its title is "Cambrian-1: A Fully Open, Vision-Centric Exploration of Multimodal LLMs". This paper is all about building such AI models which can understand the images and texts commonly known as multimodal, as its main focus is to understand the vision not only the part of the language. In sense of structure the paper is well organised as it has shown its 5 pillars. These are 5 main areas where it will do its study, not only this the paper has clearly studied every part which he has mentioned, this paper along with clear writing also explains the technical terms before using them. Whereas there is a small weakness that paper discusses a philosophical talk in the starting which explains how humans understand the world through vision, this was an interesting thing but papers like this which are entirely based on a technical depth it covers unnecessary space of paper as it could be shortened.

RESEARCH PROBLEM AND ABSTRACT

This paper has clearly stated its research problem, the problem is that most of the recent multimodal are entirely focused on improving the language efficiency, but no one has looked at vision side that how could image can be turned into the information which model can use. The authors of this paper argues that if the vision part of the model is weak then it cannot see the image even if its language skills are best. This research paper's abstract discuss on 3 things. One, a systematic study which compares many different vision encoders, It means the part of the

model which reads the image. Second, We noticed that a new benchmark named CV bench Will be used to test the vision centric questions. Third, identify a new way of connecting the language part with the vision named as SVA. The paper has actually delivered all these 3 things very well.

LITERATURE REIVIEW

When we closely looked at paper it does not have a single section of related work, as many of the papers do this. As it discusses this related work section throughout the introduction and the background parts, as well as it also includes a very large number of references which means that author has a very good knowledge of this field and they have placed their work properly comparing to existing multimodal models, vision encoders end benchmarks. But it was very complicated for a reviewer to quickly see how does d Cambrian-1 compared against the other methods, instead it should be in a proper grouped, so that it would be much more easier for the reviewer its methods and it could compare it with other studies also.

METHODOLOGY

The authors in this part has displayed strong efforts in Vision enc as they have tested a 23 different encoders under the same fair conditions. They want to make the comparison as of meaningful, instead of coping numbers and other papers that have used different setups. This thing has made this paper very trustworthy.

To understand this test vision properly, the authors of this paper I have noticed a very useful thing that most of the existing benchmarks would be solved without looking at the image this has made the purpose of testing vision useless. For this reason the authors of the paper have built a new benchmark called as CV bench, it was most importantly designed in such a way that a model must look at an image to answer correctly. Not only this they have also described that how did they built it and how did they checked the benchmark. They also included a human verification questions. They have used it for checking the correctness of the model before knowing which model would perform good or bad.

The writers of the paper have also discovered a new way of connecting the language model to the vision encoder it is known as an SVA (spatial vision aggregator). In this way, we do not just give the list of image features as it will lose the information about where the things are in the image . Usually SVA keeps the spatial part of the image and it allows the language model to repeatedly look back to the important part of the image while its generating its answer. This complete method has been explained by using the proper mathematical approach, this makes the easy for other researchers to reproduce it.

The Authors of this paper has not selected the training data randomly they ran a number of experiments, To build a good instruction tuning data set, The data which tells the model how to answer the questions. They have tested different mixes and amount of data to create their own data sets like Cambrian 7M and Cambrian 10M. The complete methodology is very reproducible format, as the authors have released their codes, model weights and their data publicly. However we have noticed that many of these experiments were running on medium sized datasets. And then it was assumed by them that this would be applied same when it is scaled up to the bigger models and bigger data. Also the authors have checked the assumptions in a few cases but not in every single experiment, it cannot be assumed that it has been fully retested at a full scale.

EVALUATION AND RESULTS

The authors of this paper has evaluated its models on more than a 20 different benchmarks which covers at different skills and then they have grouped them into a different 4 categories these categories are general understanding, knowledge based questions, text/chart reading and vision centric task. This grouping was not done randomly as they have first looked at the benchmarks which tend to each other rather than just guessing. Then the final Cambrian 1 model Was compared against the strong existing models like GPT 4V and Gemini, and the results was very surprising as the authors Cambrian 1 was performing competitively specially on the vision centric benchmark which the authors have built by themselves.

One thing I like about the report is that the authors has not entirely talked about his successes as well as they have also discussed the problems which they encountered during the process they have also discussed the " answer machine phenomenon " problem where model is trained on a short answer data so the model starts giving the answers very short, some questions need specifically an explanations and they have explained this how did they solve this problem using a better system prompts during the training. Not only these they have verified their results by checking specifically on their scoring method (it compares the simple text machine scoring with a more expensive AI judging scoring) so they can make sure that Their evaluation results does not show any misleading, and then it was reported that the difference between both the methods was too small. This kind of approach is really a trustworthy approach we can believe it, although there was some benchmark scores in this area which still carries some kind of noise in it.

CONCLUSION, DISCUSSIONVAND OPEN RESEARCH PROBLEMS

The authors of the people are very honest regarding their findings and their limitations as they have been celebrating their results as well as they have been discussing their problems which they encountered during the process. They have encountered that the vision encoders they use they still processes the image in somewhat unordered "bag of features" way , this created the model to be limited as reason about positions and relationships in the image. Additionally, they have also seen that their model was working only with a fixed image resolutions, it does not any flexibility in the resolutions, this has limited there performance on a very detailed images. Although they have

covered it by a method which they suggested for the future named as reinforcement learning this can be used to improve the response quality further. They have clearly stated these limitations without hiding them, this is the strength of this paper, this helps the other researchers to work on such a problems and give some valuable suggestions or take the study more ahead.

This paper has also included uh dedicated section which discusses the misuse of this technology for example generating the misinformation and also they have discussed some ways to reduce these risks, they have checked numerous times weather any of their tested images have generated such information or not in their training data. This kind of minor information is not always properly cited in the papers this reflects the quality and the care taken by the researchers in this study.

OVERALL ASSESSMENT

This paper is really well structured and properly explained as well as carefully executed with a proper transparency. They have shown a proper fairness of their strengths of its vision encoder comparison, the creation of their genuinely independent bench, further they explained a new connector design SVA. After all winnings and successes they have not hided there weaknesses and open problems. However we've seen that its related work section was not properly grouped it could be better organised, also the methodology which generalizes the findings on the small scale experiments where claimed and the larger scale models were judged on the basis of the small scale experiments without any testing. Finally the paper is solid and useful contribution to the research area in the field of multimodal AI models. They have publicly released their codes data and models which adds the real value to the researchers who wants to build on this work.